MILITARY SPECIFICATION

RADAR SET: AN/APS-96

This specification has been approved by the

Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - This specification contains the detailed requirements which have been established for an early warning type search radar. This equipment is specifically designed for, and constitutes a portion of the W2F-1 Carrier Based AEW/CIC System. This equipment provides target information necessary to compute range, azimuth and elevation.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Unit	Type Designation	Max. Allow. Size (Inches) W H L	Max. Allow Wt. (lbs.)	Appl. Par.
Receiver Panel	R-(*)/APS-96			3.4.1
Receiver Group AMTI Group Matched Filter Group Synchronizer Group Transmitter Frequency Converter Group Stable Local Oscillator (STALO)				
R. F. Driver	RF-(*)/APS-96			3.4.2
R. F. Amplifier (High Power)	RF-(*)/APS-96			3.4.3
Modulator (High Power)	MD-(*)/APS-96			3.4.4
Power Supply (High Voltage)	PP-(*)/APS-96			3.4.5
Transformer Assembly	TF-(*)/APS-96			3.4.6
Reflectometer	ME-(*)/APS-96			3.4.7

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Duplexer Assembly	CU-(*)/APS-96	3.4.8
Directional Coupler	CU-(*)/APS-96	3.4.9
Coaxial Switch	SA-(*)/APS-96	3.4.10
Radar Set Control	C-(*)/APS-96	3.4.11
Air Compressor	HD-(*)/APS-96	3.4.12
Dehydrator	HD-(*)/APS-96	3.4.13

* The paraentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the Bureau of Aeronautics upon application by the contractor for assignment of nomenclature in accordance with 3.2.14. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

1.3 Associated Equipment - The equipment shall be capable of operating with the following associated equipment which is not to be supplied as part of this equipment:

(1) CP-413/ASA-27 Computer - Detector

(2) AN/ASA-27 Computer - Indicator Group

(3) AN/APA-143 Antenna Group

(4) Aircraft Electronic Equipment Cooling System

(5) AN/USM-25 Test Oscilloscope

(6) Dummy Load (High Power)

(7) Dummy Load (Low Power)

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitations for bids, form a part of this specification:

SPECIFICATIONS

Military -

MIL-T-25453	Test Points and Test Facilities, Airborne and Associated Electronic Equipment
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment

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MIL-C-6781	Control Panel, Aircraft Equipment, Rack and Console Mounted
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-P-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing and Marking of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-K-25049	Knobs, Pointer
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-C-18012	Control Configuration and Markings (for Plastic Lighting Plates, Control Panels, and Placards)

Bureau of Aeronautics -

EI-207	Control Panels, Electrical and Electronic Equipment, Rack and Console Mounted, Installation of
SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

Air Force - Navy Aeronautical -

MS-25176	Shaft, Control Knob

Bureau of Aeronautics -

MS28053	Indicator, Elapsed Time

2.2 Availability of Documents -

(1) Give both the title and number when requesting specifications, standards, drawings, and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot (AD), Philadelphia 11, Pennsylvania.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Bureau of Aeronautics.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance, or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as non-standard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Non-standard parts of an electrical nature defined in Specification MIL-E-5400, must be approved before the preproduction model of the equipment is submitted for tests. The request for approval must be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Tube Types - The choice and application of electron tubes used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

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(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in the size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts or processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

3.2 Design and Construction -

3.2.1 General - This equipment shall fulfill all applicable design and construction requirements of Specification MIL-E-5400 for Class I equipment unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily and shall withstand, in a non-operating condition all the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment except as follows:

3.2.2.1 Altitude - This equipment shall perform as specified herein when operating at any cabin pressure up to and including 15,000 feet. (11.1 inches H g). The equipment shall not suffer damage as a result of a pressure change equivalent to an altitude change from 5,000 feet (24.9 inches Hg) to 40,000 feet (5.5 inches Hg) in 2 seconds time, and in 10 seconds time for a change in the reverse direction.

3.2.2.2 Temperature - Altitude - The equipment shall be capable of continuous operation over the temperature range as specified in MIL-E-5400 with maximum cabin altitude of 15,000 feet.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room Ambient (30°C ± 10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0V AC

3.2.4 Total Weight - The total weight of the equipment excluding cables and coaxial transmission lines not supplied as part of the radar set shall not exceed 1575 lbs.

3.2.5 Operational Stability - The equipment shall operate with specified performance for 24 hours, continuously or intermittently, without necessity for readjustment of any controls other than operating controls.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer, and a list of such parts shall be submitted to the Bureau of Aeronautics before start of preproduction tests.

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3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6.3 Elapsed Time Indicator - And Elapsed Time Indicator shall be provided for registering the time that the transmitter is in STANDBY and OPERATE. The indicators shall be in accordance with the Bureau of Aeronautics Drawing MS-28053.

3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external inter-unit wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring to the unit shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, external inter-unit cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Control Panels - All remote control panels shall be satisfactory for mounting in a console arrangement when used in aircraft. These units shall meet the requirements of Specification MIL-C-6781 except that the requirements for a transilluminated plastic plate and internal red lighting shall not apply. A plastic plate of grey color satisfactory for use with polarized white lighting shall be provided. These units shall also be equipped with fasteners conforming to Specification MIL-F-25173 for mounting in accordance with Specification El-207. Knobs conforming to Specification MIL-K-25049 shall be used. Push buttons shall have black legends on diffused white background and the sides shall be opaque (non-light transmitting). The arrangement of controls and all lettering and markings proposed for use on the front panel of control units shall comply with MIL-C-18012 and shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 with characteristics as defined therein and shall give specified performance from the following power sources as specified hereinafter. The power required shall not exceed the specified amounts.

AC Power (3 Phase) -

(1) 115/200 Volt
3 Phase Operating
Voltage Limits - - - - - - - - 108.0 to 121.0 Volts

(2) Operating Frequency Limits - 380 to 420 CPS

3.2.11 Radar Load Requirements - The unbalanced load per phase shall not exceed 30% of the total 3 phase power or 500 VA whichever is smaller. Three phase power shall not exceed 18 KVA and the power factor shall be greater than 0.85.

3.2.12 Primary Power Switching - There shall be no primary power protection or switching in this equipment. This protection shall be supplied at the aircraft distribution box. The protective devises characteristic for this equipment shall be supplied as design data specified in 4.2.2.1 and in the El specification for this equipment.

3.2.12.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

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3.2.12.2 Overvoltage Protection - Overvoltage protection, other than primary power for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of protective devices shall be held to a minimum. There shall be no fuses in the equipment.

3.2.13 Standby Provisions - The equipment shall provide for standby operation wherein operating voltage, filament and plate voltages are applied to all tubes with the exception that plate voltage shall not be applied to the modulators.

3.2.14 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.15 Finishes - Finishes shall meet the applicable requirements except that color coding shall be permissible.

3.2.16 Interchangeability - The interchangeability requirements between like assemblies, subassemblies, and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that the requirements of Specification MIL-E-5400 conflict with requirements for interchangeability, the interchangeability requirements shall govern.

3.2.17 Relays - All relays used in this equipment shall be of the plug-in variety and shall be hermetically sealed.

3.2.18 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance and shall meet the requirements of Specification MIL-T-25453.

3.2.19 Maintenance Controls - All maintenance controls except those tuning RF coils shall be in accordance with the following:

(1) The contractor shall furnish in writing justification for each maintenance control.

(2) The jacks required to make maintenance tests and adjustments shall be provided.

(3) The controls and test jacks shall be accessible from the front of the unit without removal of the unit from its mounting provisions; however, hinged access doors or removable front panels are permissible.

(4) The controls and jacks shall be labelled as to function. Associated jacks and controls shall be clearly identified.

3.2.20 Radio Interference Control - The generation of radio interference by the equipment and vulnerability of the equipment to radio interference shall be controlled in accordance to Specification MIL-I-6181; however, the upper frequency limit shall be extended from 1000 MC to 14 KMC.

3.2.21 Marking of Connectors - Receptacles in this system shall be marked where feasible with appropriate names or abbreviations indicating other units that the connector is normally connected to or it shall be marked with the function.

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3.3 Performance -

3.3.1 General - The Radar Set shall incorporate the following techniques:

(1) Matched Filter Pulse Compression

(2) Time Average Clutter Coherent Moving Target Indication (TACCAR)

(3) Sensitivity Time Control (STC)

3.3.2 Warm-Up Time - The warm-up time for the equipment shall not exceed 15 minutes for temperature above 0°C and 30 minutes if the ambient starting temperature is less than 0°C.

3.3.3 Frequency Characteristics - This equipment shall be capable of operating at any of 10 selectable frequencies within the 406 to 450 megacycle frequency range. By automatic tuning it shall be possible to select any of these 10 frequencies. The automatic frequency selector system may have maintenance adjustment provisions.

3.3.4 Inputs -

(1) Primary power per 3.2.10.

(2) Received signals from the AN/APA-143 Rotodome Antenna Assembly on 50 ohm 3-1/8 inch coaxial line.

3.3.5 Radar Outputs -

(1) Transmitter Output Pulse

(2) Radar Trigger

(3) Video

(4) Radar Normal/Emergency Mode Signals

3.3.5.1 Radar Power Output - The equipment shall be capable of delivering approximately 1 megawatt peak power over the 406 to 450 megacycle range at a minimum of 3800 watts average power. A design value of 3 megawatts peak power shall be used for all high power radio frequency components terminated into a flat load. The transmitted output pulse shall have an approximately rectangular shape of 12.8 ± 0.3 microseconds duration at a repetition frequency of 300 pulses per second.

3.3.5.2 Radar System Trigger Outputs - The Radar Set shall supply the following triggers for other electronic equipment in the W2F aircraft.

(1) IFF "B" Trigger: This trigger shall normally be coincident with the PRF trigger from the AMTI group. The PRF shall be 300 ±5 with less than .01 microsecond jitter and shall occur 430 ± 0.1 microsecond before the system trigger described in (4).

(2) IFF "A" Trigger: This trigger shall occur 70 ± 0.1 microseconds before the system trigger and shall have less than .01 microsecond jitter.

(3) Blanking Trigger: This trigger shall occur 30 ± 5 microseconds ahead of the system trigger and shall be 2.0 ± 0.5 microseconds wide at the 50% amplitude level with less than 1.0 microsecond jitter.

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(4) System Trigger: This trigger shall be coincident within ± 0.1 microsecond with the compressed video return for a target at zero range.

3.3.5.2.1 Radar System Trigger Characteristics - The triggers listed above shall have the following characteristics unless otherwise specified:

(1) Pulse width shall be 1 ± 0.5 microseconds at the 50% amplitude level.

(2) Pulse amplitude shall be 25 ± 5 volts peak AC coupled into a load resistance of 100 ohms with a shunt capacity not to exceed 500 micro-microfarads.

(3) Radar PRF shall be 300 ± 5 cycles per second.

(4) Pulse to pulse leading edge jitter shall not be more than 0.03 microseconds.

(5) The rise time shall not be greater than 0.05 microseconds from the 10% to the 90% amplitude points.

(6) The polarity shall be positive.

3.3.5.3 Radar System Video Outputs - With receiving system input signals adequately simulating the radar transmitter output and the expected R.F. returns from the targets, the sea and the land with power as great as 100 db above noise, the radar system video outputs shall be as follows:

3.3.5.3.1 Video Outputs to AN/ASA-27 - Computer - Indicator - The following types of video shall be available for use by the AN/ASA-27 Computer-Indicator:

(1) Matched Filter Video

(2) Matched Filter-AMTI Video

(3) Long Pulse Video

(4) Long Pulse AMTI Video

3.3.5.3.1.1 Characteristics of Video Outputs to Computer - Indicator - The video types listed above shall have the following characteristics:

(1) 2 ± .5 volts peak and .6 ± 0.15 volts RMS thermal noise for a point target that is not limited.

(2) The compressed pulse shall not be wider than 0.3 microsecond at the 50% voltage amplitude level.

(3) The rise time shall not be greater than 0.3 microsecond from 10% to 90% amplitude points.

(4) The compressed side lobes (in range) shall be more than 18 db below the maximum pulse amplitude.

3.3.5.3.2 Video Outputs to CP-413/ASA-27 Computer-Detector - The following types of video shall be available for use by the CP-413/ASA-27 Computer-Detector:

(1) Matched Filter Video

(2) Matched Filter AMTI Video

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3.3.5.3.2.1 Characteristics of Video Outputs to Computer-Detector - The video type listed above shall have the following characteristics:

(1) 6 volts peak maximum and 0.6 ± 0.15 volts RMS thermal noise.

(2) The compressed pulse shall not be wider than 0.3 microsecond at the 50% voltage amplitude level.

(3) The compressed pulse side lobes (in range) shall be more than 18 db below the maximum limited pulse amplitude.

(4) The rise time shall not be greater than 0.3 microseconds from 10% to 90% amplitude points.

3.3.5.3.3 Dual Pulse Resolution - The receiving system shall be capable of resolving 2 input pulses, during Matched Filter operation, that are displaced 0.4 microseconds.

3.3.5.3.4 Cancellation Ratio - The cancellation ratio shall be at least 25 db during Matched Filter operation with a .2 microsecond pulse.

3.3.5.3.5 Video Bandwidth - The overall video response shall be such that pulses from 0.2 to 1000 microseconds shall have a maximum rise time of 0.05 and a drop of not more than 1.0 db.

3.3.5.3.6 Video Noise Level - The video noise level shall be at least 10 db below the receiver thermal noise level.

3.3.5.3.7 Video Amplifier Linearity - The video amplifier linearity shall be within ± 25% from 5% to 150% of the normal maximum output. This shall be true on a 10 microseconds signal following a 1000 microseconds maximum amplitude signal by 10 microseconds.

3.3.5.4 Radar Normal/Emergency Signals - The Radar Set shall provide a ground for the Computer-Detector to signal that the Radar Set is in an emergency mode of operation.

3.3.6 Overall VSWR - The radar equipment shall be capable of operating into a transmission line of characteristic impedance equal to 50 ohms with a Voltage Standing Wave Ratio of 1.3.

3.3.7 Group Failure Indicators - The following groups shall include go-no-go type test devices.

(1) Receiver Group

(2) Matched Filter Group

(3) AMTI Group

(4) Synchronizer Group

3.3.8 Pressurization - All pressurized components shall be capable of withstanding not less than 5 psi in excess of maximum operating pressure.

3.3.9 Receiving Performance -

3.3.9.1 Noise Figure - The overall noise figure of the system shall not exceed 6.5 db.

3.3.9.2 Minimum Discernable Signal - The minimum discernable signal shall be -118 dbm during long pulse operation and -117 dbm during Matched Filter operation.

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3.3.9.3 Spurious Response - The response at off-frequency signals shall be as follows:

(1) at least 20 db down at 10 mc off center frequency

(2) at least 50 db down at 20 mc off center frequency

(3) at least 70 db down at 40 mc off center frequency

3.3.10 Pulse Stability - The total pulse-to-pulse jitter in the transmitted pulse shall not exceed .01 microsecond.

3.3.11 System AMTI Performance - The system AMTI cancellation ratio shall be not less than 25 db at 200 nautical miles and shall include the transmitter and receiver components in the normal and emergency modes.

3.3.12 Cooling - Cooling as required shall be accomplished by air and by heat transfer to liquid coolant; however, liquid cooling shall be held to an absolute minimum. Where liquid cooling is required, suitable connectors consistant with the W2F-1 liquid cooling system shall be provided. The coolant pump assembly, interconnecting tubing and their fittings shall not be supplied as part of this equipment.

3.4 Detail Requirements -

3.4.1 Receiver Panel - R-(*)/APS-96 - The Receiver Panel shall consist of the following groups:

(1) Receiver Group

(2) AMTI Group

(3) Matched Filter Group

(4) Synchronizer Group

(5) Transmitter Frequency Converter Group

(6) Stable Local Oscillator Group

3.4.1.1 Receiver Group -

3.4.1.1.1 Function - The Receiver shall be of the superheterodyne type, capable of receiving radio frequencies in the band from 406 to 450 mc, from the radar duplexer assembly. The Receiver shall invert the spectrum of the received signals in conversion to intermediate frequency. The output shall be as specified in paragraph 3.4.1.1.13.

3.4.1.1.2 Receiver Performance - The receiver components shall have an overall noise figure including R.F. amplifier, mixer, and I.F. pre-amplifier of not more than 5.5 db for any operating frequency in the 406 to 450 megacycles per second region. Receiver characteristics shall include STC, sufficient gain to produce 1.0 volt peak-to-peak thermal noise output across 93 ohms to be supplied to the matched filter. A signal at a range of 10 miles and 80 db above thermal noise shall not produce limiting at the output of the receiver second mixer.

3.4.1.1.3 R.F. Amplifier - The Receiver shall include an R.F. amplifier that shall have a gain of 20 db minimum at 425 mc with a bandwidth of 60 mc minimum at the 3 db point.

3.4.1.1.4 Receiver First Mixer - A mixer shall be provided for mixing the received amplified R.F. frequency and that generated by the STALO. No mixer adjustments shall be necessary in the 406 to 450 megacycles band for proper operation.

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3.4.1.1.5 Receiver First I.F. Amplifier - The amplifier shall be centered at 122.5 mc which is the difference between the STALO and the received R.F. signal. Its bandwidth shall be at least 15 mc between the 3 db points and shall have a gain of 20 db minimum, including the receiver second mixer. This IF strip shall have STC circuitry which produces -25 ± 5 db of sensitivity control at 10 N.Mi., approximately -3 db sensitivity control at 50 N.Mi., and recovers within .5 db of normal sensitivity at 80 N.Mi.

3.4.1.1.6 Local Oscillator - A crystal controlled local oscillator signal of sufficient output shall be provided at 152.5 mc to provide efficient conversion in the second mixer.

3.4.1.1.7 Receiver Second Mixer - A mixer shall be provided for mixing the 122.5 mc intermediate frequency signal with the local oscillator signal to produce a frequency spectrum inversion of the R.F. pulse centered at 30 mc.

3.4.1.1.8 Receiver Second IF Limiting Amplifier (Wide Band) - A limiting type IF amplifier shall be provided having a 10 mc bandwidth minimum centered at 30 mc and a gain of 80 db minimum. A maintenance gain control shall be provided.

3.4.1.1.9 Limiting IF Amplifier (Narrow Band) - A limiting type, narrow band IF amplifier shall accept signals from the second mixer and provide output IF signals in the emergency mode of operation suitable for the AMTI. This amplifier shall have a bandwidth of 125 ±25 kilocycles centered at 30 megacycles. The gain of the amplifier shall be approximately 85 db.

3.4.1.1.10 IAGC IF Amplifier (Narrow Band) - An IF amplifier including fixed IAGC shall accept signals from the second mixer and provide raw output video signals, in the normal and emergency mode (see paragraph 3.3.5.3.1) suitable for the AN/ASA-27 Computer-Indicator. This amplifier shall have a bandwidth of 125 ± 25 kilocycles centered at 30 megacycles. The gain shall be approximately 85 db and shall be controlled by an IAGC IF gain control located on the Radar Set Control.

3.4.1.1.11 Radar Automatic Evaluator - A test pulse shall be derived from the transmitter RF pulse by means of a delay line circuit for the purpose of checking the alignment and stabilities of the radar system. This test device shall be capable of indicating a 3 db degradation of system performance, and shall be operable from the Radar Set Control.

3.4.1.1.12 Receiver Inputs - The following are the inputs to the Receiver.

(1) Radar return signals from Duplexer through 50 ohm coax.

(2) Any of 10 selectable STALO frequencies.

(3) Radar pretrigger described in paragraph 3.4.1.4.5.

(4) Transmitted pulse sample for the Radar Automatic Evaluator.

3.4.1.1.13 Receiver Outputs - The following are the outputs of the Receiver.

(1) 30 megacycle wide band IF signals limited on thermal noise to 1 volt peak-to-peak for the Matched Filter.

(2) 30 megacycles long pulse IF signals for emergency operation with AMTI.

(3) 30 megacycles long pulse video signals during normal and emergency operation for the AN/ASA-27 Computer-Indicator.

3.4.1.2 Matched Filter Group -

3.4.1.2.1 Function - The Matched Filter shall generate a 0.2 microsecond 30 mc IF pulse synchronized by the radar trigger described in paragraph 3.4.1.4.5. This pulse shall be converted to 12.8 ± .3 microseconds by a dispersive network and shall be furnished to the Transmitter Frequency

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Converter group. The 30 mc IF signals shall be accepted from the Receiver, compressed and furnished to the AMTI unit. The IF output shall be 1.0 volt peak-to-peak at 93 ohms. Raw compressed video shall be made available to the AN/ASA-27 Computer-Indicator. Raw compressed video shall also be made available to the CP-413/ASA-27 Computer-Detector.

3.4.1.2.2 Pulse Generator - The 0.2 microsecond pulse generator shall be locked to a radar trigger and shall generate an IF pulse at a frequency of 30 megacycles. The output of the pulse generator shall drive the dispersive network.

3.4.1.2.3 Dispersive Network - The dispersive network shall be used during transmission and reception, at an IF frequency of 30 Mc. It shall have a delay vs. frequency characteristic over a 5 megacycle bandwidth of the pulse spectrum such that pulse stretching occurs on transmit and pulse compression occurs on receive. The compressed pulse shall have a rise time of 0.3 microseconds maximum measured between the 10% and 90% level.

3.4.1.2.4 Gate Circuits - Time gates shall be provided in the Matched Filter to switch between transmit and receive.

3.4.1.2.5 Matched Filter Inputs -

(1) 30 megacycle 1 volt peak-to-peak limited IF signals from the receiver wide band IF amplifier.

(2) Radar trigger as described in paragraph 3.4.1.4.5 for .2 microsecond pulse generator.

(3) Radar pretrigger as described in paragraph 3.4.1.4.5 for gating.

3.4.1.2.6 Matched Filter Outputs -

(1) 30 megacycle 1 volt peak-to-peak IF signals for AMTI.

(2) Raw compressed video signals as described in paragraph 3.3.5.3.2 for the CP-413/ASA-27 Computer-Detector.

(3) Raw compressed video signals as described in paragraph 3.3.5.3.1 for the AN/ASA-27 Computer-Indicator.

(4) 12.8 microsecond, 30 megacycle expanded pulse for the Transmitter Frequency Converter Group.

(5) 30 megacycle IF transmitter sample pulse for phase locking the Coho in the AMTI group.

3.4.1.3 AMTI Group -

3.4.1.3.1 Function - The AMTI unit shall receive IF signals from the Matched Filter unit and shall operate on these signals as necessary to provide cancelled video for the Computer-Detector and the Computer-Indicator. Own ships motion and clutter motion effects shall be reduced from the return signal by time averaging the AMTI phase reference over at least one pulse repetition period. Phase comparison cancellation shall be used including amplitude limiting of the signal in the clutter region with provision for matching the amplitude levels of clutter residue and thermal noise beyond the clutter region. The AMTI unit shall also provide a 300 cps reference trigger to the synchronizer.

3.4.1.3.2 Delay Line - A quartz delay line capable of providing a 3333 ± 30 microsecond delay shall be used.

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3.4.1.3.3 Video Test Pulse Generator - A video test pulse shall be generated to check cancellation ratio as a maintenance feature. Means shall be provided for this pulse to be available either every sweep or every other sweep on a selected basis, in order that both fixed and moving targets may be simulated for test purposes.

3.4.1.3.4 AMTI Inputs -

(1) 30 megacycles 1 volt peak to peak IF signals from Matched Filter.

(2) 30 megacycles 1 volt peak to peak IF signals from the limiting narrow band IF amplifier.

(3) 30 megacycle IF transmitter sample pulse from Matched Filter.

3.4.1.3.5 AMTI Outputs -

(1) Video signals as described in paragraph 3.3.5.3.2 for CP-413/ASA-27 Computer-Detector.

(2) Video output as described in paragraph 3.3.5.3.1 for AN/ASA-27 Computer-Indicator.

(3) Reference trigger shall be supplied to the Synchronizer and shall not have jitter, pulse to pulse, in excess of 0.01 microseconds.

3.4.1.4 Synchronizer Group -

3.4.1.4.1 Function - The Synchronizer portion shall be capable of receiving a reference trigger from the AMTI unit, and generating all the required triggers as defined in this specification. The Synchronizer shall also be capable of generating an emergency trigger in the event of loss of the reference trigger in the AMTI unit.

3.4.1.4.2 Delay Line - Delay line circuitry shall be provided to furnish triggers as shown in Figure 1.

3.4.1.4.3 Emergency Repetition Rate Generator - The emergency repetition generator shall generate 300 ± 5 pulses per second. This generator shall be put in operation manually.

3.4.1.4.4. Synchronizer Inputs - The Synchronizer shall accept a reference trigger from the AMTI group described in paragraph 3.4.1.3.5 (3).

3.4.1.4.5 Synchronizer Outputs - The Synchronizer shall provide the following outputs:

(1) System Trigger

(2) Radar Trigger

(3) Radar Pretrigger

(4) Blanking Trigger

(5) IFF "A" Trigger

(6) IFF "B" Trigger

Triggers (1), (4), (5), (6) shall be as described in paragraph 3.3.5.2. Triggers (2), (3) shall have characteristics as required for proper Radar Set operation. The trigger timing shall be as indicated in Figure 1.

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3.4.1.5 Transmitter Frequency Converter -

3.4.1.5.1 Function - The Transmitter Frequency Converter shall accept a 30 megacycle 1 volt peak-to-peak expanded pulse at 93 ohms from the Matched Filter and convert this pulse to 406 to 450 megacycles at 50 milliwatts for the Medium Power RF Amplifier. During emergency operation a 12.8 microsecond narrow band pulse shall be generated in this group and converted to the same channel frequency and power as for normal operation for the Medium Power RF Amplifier.

3.4.1.5.2 Transmitter Local Oscillator - A crystal controlled local oscillator shall be provided at 92.5 mc with sufficient output to provide efficient mixing in the first transmitter mixer.

3.4.1.5.3 First Transmitter Mixer - A balanced mixer shall be provided for mixing the output signal from the matched filter with the local oscillator signal from the STALO to generate the 122.5 mc intermediate frequency. The mixer shall have a bandwidth of 10 mc minimum at the 3 db points.

3.4.1.5.4 Transmitter Intermediate Frequency Amplifier - This amplifier shall have a bandwidth of at least 15 mc and sufficient gain at 122.5 mc to provide efficient mixing in the second mixer.

3.4.1.5.5 Second Transmitter Mixer - A balanced mixer shall be provided for mixing the output signal from the IF amplifier with one of the ten STALO frequencies. The amplitude of the unwanted harmonics generated in the mixer shall be kept to a minimum. No mixer adjustments shall be necessary in the band 406 to 450 megacycles for proper operation.

3.4.1.5.6 Transmitter-Frequency Converter Inputs - This group shall accept the following inputs:

(1) 12.8 microsecond, 30 megacycles expanded pulse from the Matched Filter.

(2) Any of 10 selectable STALO frequencies.

(3) Radar Trigger.

3.4.1.5.7 Transmitter-Frequency Converter Outputs - The outputs of this group shall be any one of 10 frequencies in the 406 to 450 megacycles band.

3.4.1.6 STALO -

3.4.1.6.1 Function - The STALO shall provide any of 10 selectable stalo frequencies to facilitate efficient mixing in the first receiver mixer and the second transmitter mixer. These frequencies shall be selected by means of a frequency selector switch located on the Radar Set Control. Two outputs will be provided, one to the second transmitter mixer, located in the Transmitter Frequency Converter, and another to the receiver first mixer, located in the Receiver. The STALO frequencies shall be 122.5 Mc above the transmitted frequency.

3.4.2 R.F. Driver - RF-(*)/APS-96 - The R.F. Driver shall consist of a medium power modulator and a medium power R.F. amplifier with the following characteristics:

3.4.2.1 Modulator (Medium Power) - This assembly shall accept the high voltage DC from a power supply, convert it to a high voltage driving pulse for the RF amplifier (medium power).

3.4.2.2 Radio Frequency Amplifier (Medium Power) - This assembly shall accept RF pulses from the Transmitter-Frequency converter and amplify these pulses to drive the RF Amplifier (High Power) and shall perform as follows:

(1) The output shall be coupled to the RF Amplifier (High Power). The output power shall be not less than 10 kilowatts peak when coupled to a 50 ohm line having a VSWR not greater than 1.25.

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(2) The output amplifier shall be automatically tuned to one of the 10 preset channels. Each channel shall have a minimum bandwidth of fifteen megacycles between the 3 db points.

(3) The input shall be fixed tuned and shall have minimum bandwidth of fifty megacycles between the 3 db points.

(4) The envelope width of the RF pulse shall be 12.8 ± .3 microseconds measured 3 db down from the peak.

3.4.2.3 R.F. Driver Inputs - The R.F. Amplifier shall accept the following inputs:

(1) A 50 milliwatt R.F. pulse from the Transmitter Frequency Converter in the 406 to 450 megacycle band.

(2) A Radar pretrigger as indicated in paragraph 3.4.1.4.5.

3.4.2.4 R.F. Driver Outputs - The R.F. Amplifier shall provide the following outputs:

(1) A trigger pulse for the Modulator (High Power).

(2) An R.F. pulse of not less than 10 kilowatts peak for the R.F. Amplifier (High Power).

3.4.3 Radio Frequency Amplifier (High Power) - RF-(*)/APS-96 - This unit shall accept a high voltage pulse from the Modulator (High Power) and a medium power radio frequency pulse from the R.F. Driver and generate a high power R.F. pulse in the R.F. amplifier tube which will deliver the high power R.F. pulse to the R.F. transmission line. This unit shall have the following characteristics:

(1) The power tube shall be selected on the basis of stability, power handling capabilities, efficiency, reliability and duration of operating life.

(2) The R.F. Amplifier (High Power) shall be automatically tuned to one of 10 preset channels. The automatic tuner shall be controlled from the Radar Set Control.

(3) The 3 db bandwidth of each channel shall be not less than fifteen megacycles per second.

3.4.3.1 Pulse Shape - The RF pulse width shall be 12.8 ± 0.3 microseconds measured 3 db down from the peak.

3.4.3.2 Cooling - Liquid cooling provisions as required shall be made in accordance with paragraph 3.3.12.

3.4.3.3 Radio Frequency Amplifier (High Power) Inputs: -

(1) 10 kilowatt peak R.F. pulse from the R.F. Driver.

(2) High Voltage pulse from the Modulator (High Power).

3.4.3.4 Radio Frequency Amplifier (High Power) Outputs - This unit shall provide R.F. pulses approximately 12.8 microseconds long at a peak power of approximately 1 megawatt as described in paragraph 3.3.5.1.

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3.4.4 Modulator (High Power) - MD-(*)/APS-96 - This unit shall accept the high voltage DC from the Power Supply (High Voltage) and a triggering pulse from the R.F. Driver and to convert them to a high voltage driving pulse for the R.F. Amplifier (High Power). This Modulator shall deliver a voltage pulse of a width of approximately 13 microseconds and of an amplitude as required to obtain approximately one megawatt peak power output from the R.F. Amplifier (High Power).

3.4.4.1 Cooling - Liquid cooling provisions as required shall be made in accordance with paragraph 3.3.12.

3.4.5 Power Supply (High Voltage) - PP-(*)/APS-96 - This unit shall rectify three phase voltage output from the Transformer Assembly, and supply the required DC voltages to the modulators. The DC output shall be controllable as required by the modulator units.

3.4.5.1 Cooling - Liquid cooling provisions as required shall be made in accordance with paragraph 3.3.12.

3.4.6 Transformer Assembly - TF-(*)/APS-96 - This unit shall provide control both automatic and remote, of the three phase voltages required by the Power Supply (High Voltage).

3.4.7 Reflectometer - ME-(*)/APS-96 - This unit shall present a visual indication of the transmitter power output and the VSWR of the transmission system. The VSWR introduced into the transmission line shall be less than 1.02. This unit shall have the following calibration:

(1) This unit shall be capable of presenting a VSWR indication to a maximum VSWR of 3 with an accuracy of ten percent.

(2) This unit shall be capable of presenting a power indication to maximum power of three megawatts with an accuracy of ten percent.

3.4.8 Duplexer Assembly - CU-(*)/APS-96 - The Duplexer Assembly shall be a broadband device for transmission and reception of radar signals in the 406 to 450 megacycles per second region. It shall include broadband TR tubes and be of rigid coaxial construction. The bandwidth measured between the 3 db points shall not be less than 60 megacycles. During the period that the transmitter is on, there shall be at least 60 db isolation between the transmitter output and the input to the receiver. The insertion loss of the duplexer on receive shall not exceed 1.0 db. The Duplexer shall be capable of carrying at least 3 megawatts of transmitted peak power when pressurized and terminated in a flat load. The high level voltage standing wave ratio shall be less than 1.15. The Duplexer shall be capable of withstanding an internal pressure of 35 psi gage.

3.4.9 Directional Coupler - CU-(*)/APS-96 - The Directional Coupler shall provide for coupling of a fixed percentage of power between the transmission line and test jacks to permit measurement of receiver sensitivity. Coupling of the power shall be attenuated approximately 42 ± 2 db, calibrated versus frequency to .5 db. The calibration shall be affixed to the Directional Coupler. The directivity shall be not less than 28 db. The VSWR introduced into the transmission line shall be less than 1.02 independent of the termination of the test jacks. The VSWR looking into the test jacks shall be less than 1.5.

3.4.10 COAX Switch - SA-(*)APS-96- This unit shall switch the high power RF energy either to the radar antenna or to a dummy load. It shall operate with RF power off. The switch shall permit operation at a peak power level of 3 megawatts. The VSWR shall not exceed 1.02 from 406 to 450 megacycles per second. With the radar transmitter connected to the dummy load, the isolation between the input and the antenna connection shall be at least 50 db.

3.4.11 Radar Set Control - C-(*)/APS-96 - This unit shall provide remote control of the Radar Set and means of monitoring its performance.

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3.4.11.1 Controls - The following controls shall be considered for inclusion on the front panel of the Radar Set Control:

(1) IAGC IF Gain Control

(2) AMTI Clutter Residue to Thermal Noise Balance Control

(3) AMTI Noise Control

(4) Antenna Start, Stop, Switching

(5) Radar Power Switching, OFF-STANDBY-OPERATE

(6) Frequency Selection

(7) Overload Reset

(8) Transmitter Power Control

(9) Mode Selection (Normal and Emergency)

3.4.11.2 Monitor - Provisions shall be made on the front panel of the Radar Set Control for monitoring operating conditions of the following types:

(1) Critical voltages and currents

(2) Pressures and temperatures

(3) Performance testing

3.4.12 Compressor - HD-(*)/APS-96 - An Air Compressor capable of supplying a minimum of 2 atmospheres absolute air pressure as required shall be provided.

3.4.13 Dehydrator - HD-(*)/APS-96 - The dehydrator shall contain an activated dessicant for maintaining dry air in the pressurized components of the system. The dehydrator shall be so constructed that, when filled with dessicant, and when the inlet and outlet are sealed, the dessicant will remain activated for a period of not less than 90 days.

3.5 Workmanship - Workmanship, fabrication, and manufacturing processes shall conform to specification MIL-E-5400.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements.

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

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(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-E-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include interference tests in accordance with Specification MIL-I-6181 and environmental tests in accordance with Specification MIL-T-5422 except that the following tests shall not be performed as part of the contractor's tests.

(1) Salt Spray

(2) Sand and Dust

(3) Explosive Atmosphere

(4) Fungus

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of

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Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subject to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 12 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 12-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 12-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 12-hour period specifiedabove may be composed of four 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 60 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling - in each of 3 planes)

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(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted on a single equipment representative of the current production. During the test, complete time and maintenance records, indicating all failures and corrective action taken shall be made. This test may be performed concurrently with the flight evaluation of the equipment which has been selected by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions -The life test may be conducted under the following simulated service conditions.

Temperature	Normal ground
Altitude	Normal ground
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol-number

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(3) The part function

(4) Name of the manufacture

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's, and EMB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

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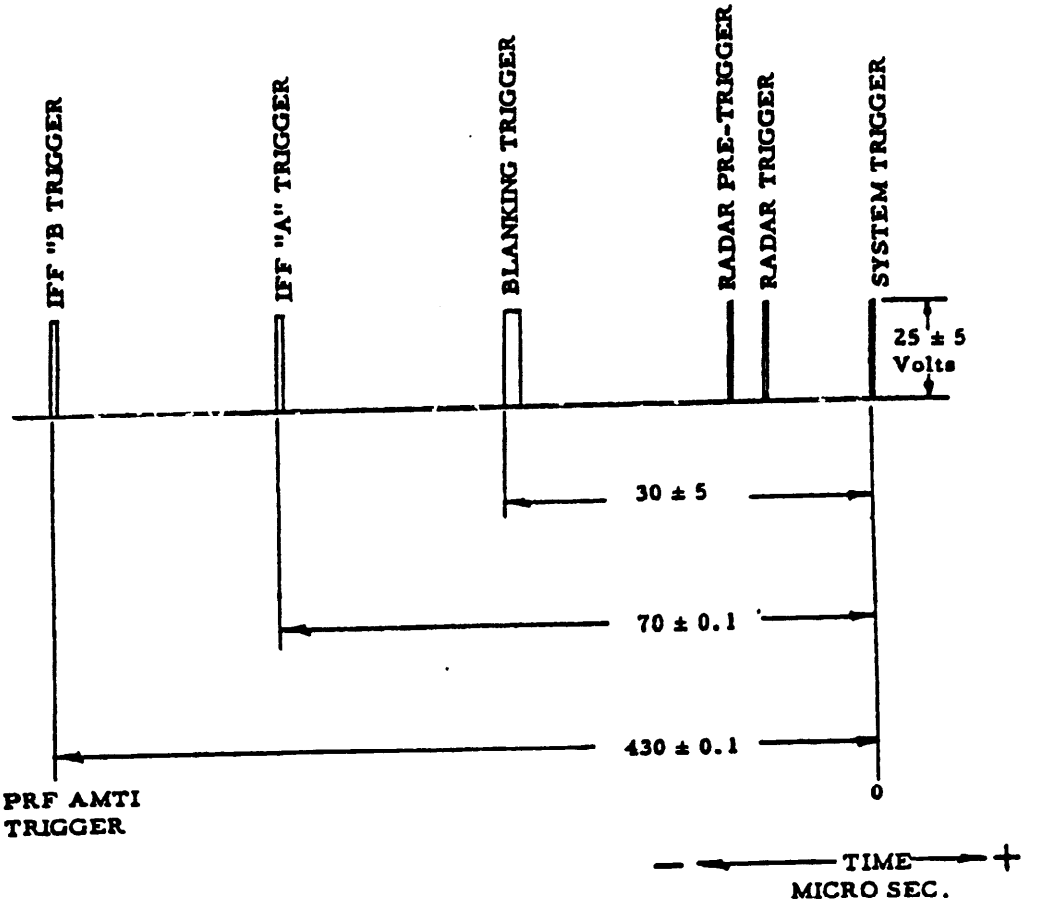
6. DEFINITIONS

6.1 Zero Range Time - Zero range time is defined as the time at which an echo from a target at zero distance from the radar set will appear as a compressed output from the Matched Filter Group.

6.2 Normal Operation Mode - Normal operation mode is defined as Radar Set operation utilizing the Matched Filter Group on both transmission and reception.

6.3 Emergency Operation Mode - Emergency operation mode is defined as Radar Set operation not utilizing the Matched Filter Group for transmission or reception but instead transmitting and receiving a conventional type of pulse modulated continuous wave whose duration is approximately equal to that of the expanded pulse during normal operation mode.

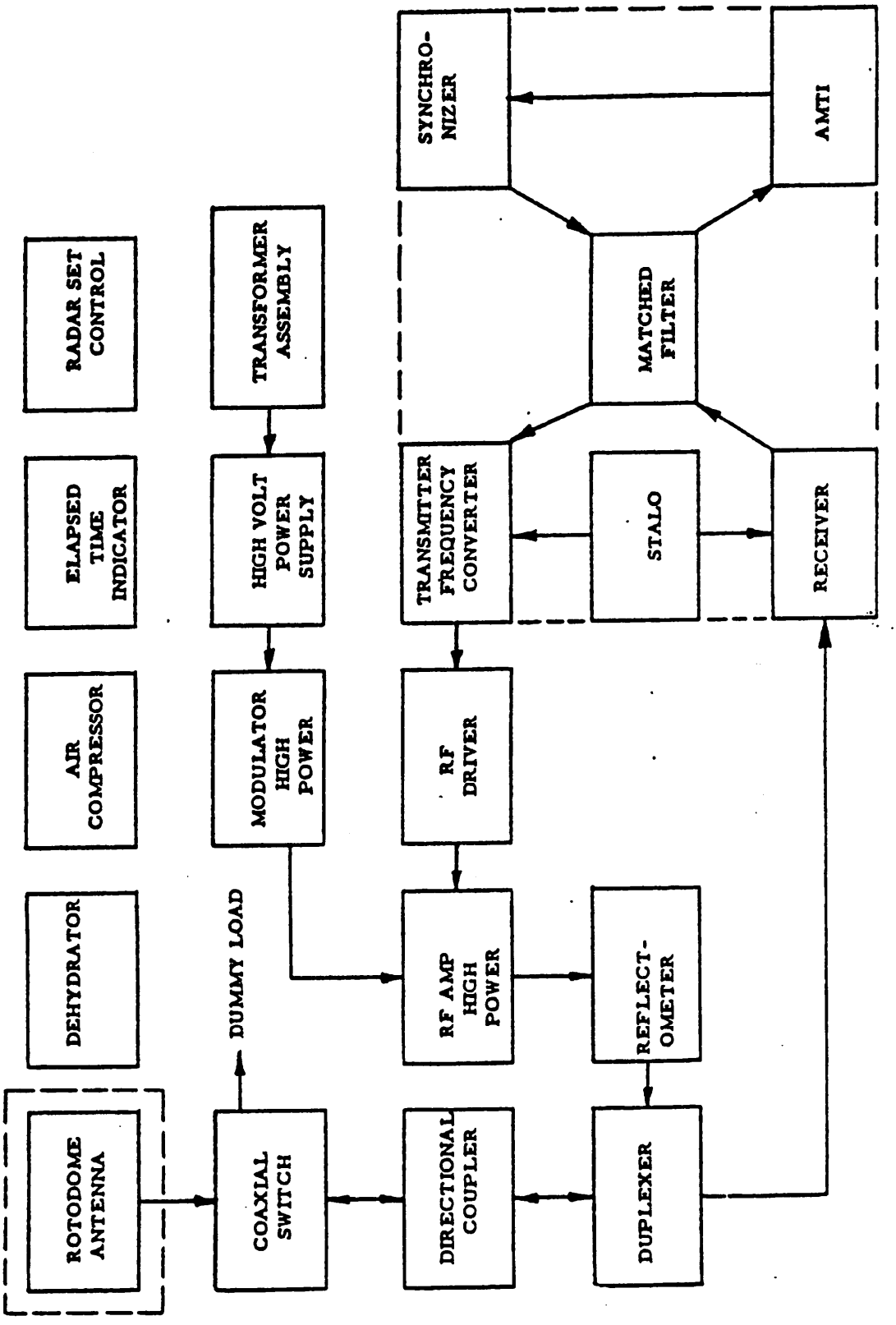
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:27Z
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TRIGGER TIMING DIAGRAM

FIGURE 1

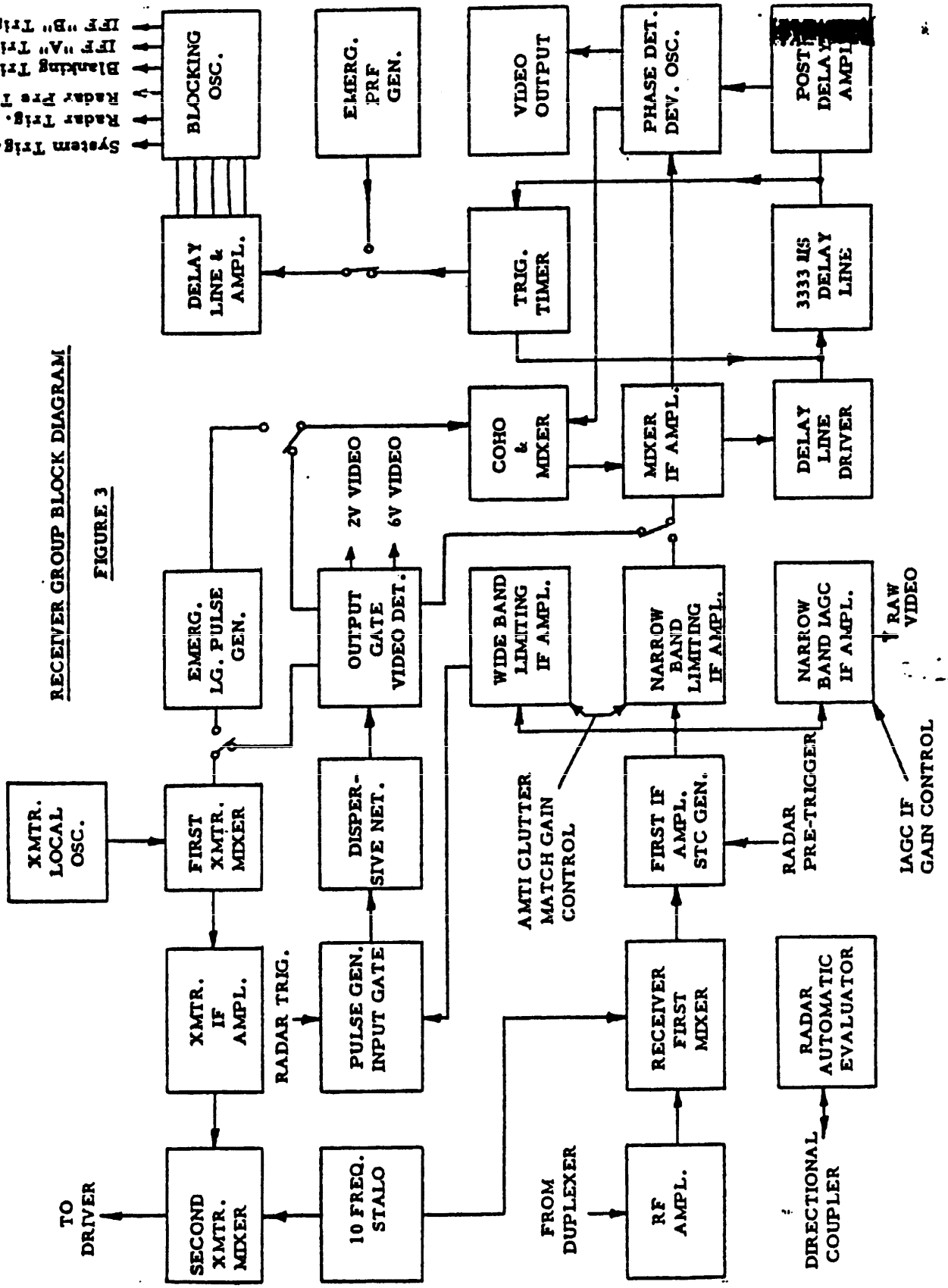
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W2F-1 RADAR BLOCK DIAGRAM

FIGURE 2

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RECEIVER GROUP BLOCK DIAGRAM

FIGURE 3

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